

Module – 1 Introduction to Data Mining

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Necessity Is the Mother of Invention

- Data explosion problem
 - Automated data collection tools and mature database technology lead to tremendous amounts of data accumulated and/or to be analyzed in databases, data warehouses, and other information repositories
- We are drowning in data, but starving for knowledge!
- Solution:
 - Data warehousing and on-line analytical processing
 - Mining interesting knowledge (rules, regularities, patterns, constraints) from data in large databases

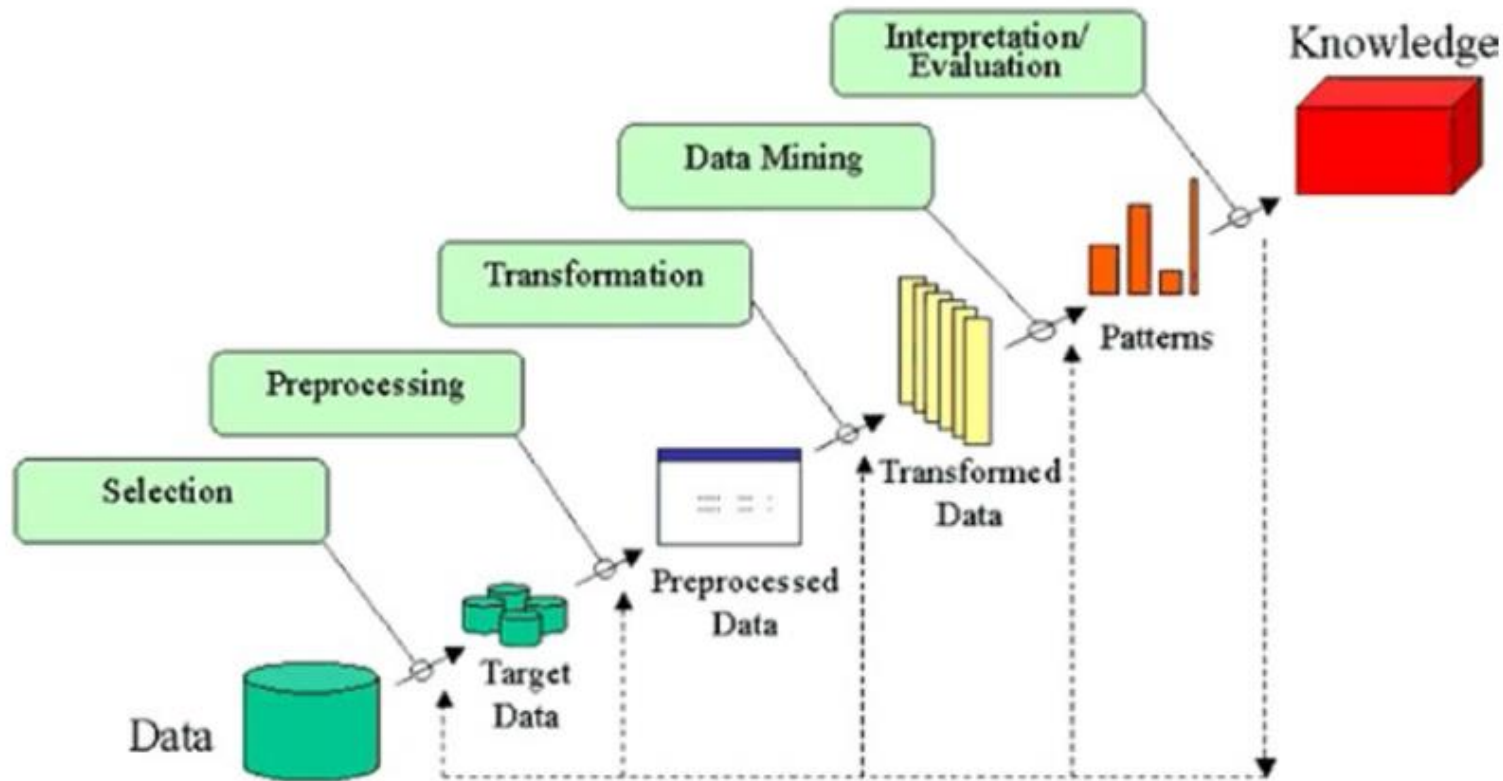
Data Mining

- It is an interdisciplinary subfield of computer science.
- It is the computational process of discovering patterns in large data sets (from data warehouse) involving methods at the intersection of artificial intelligence, machine learning, statistics, and database systems.
- Convergence of multiple disciplines

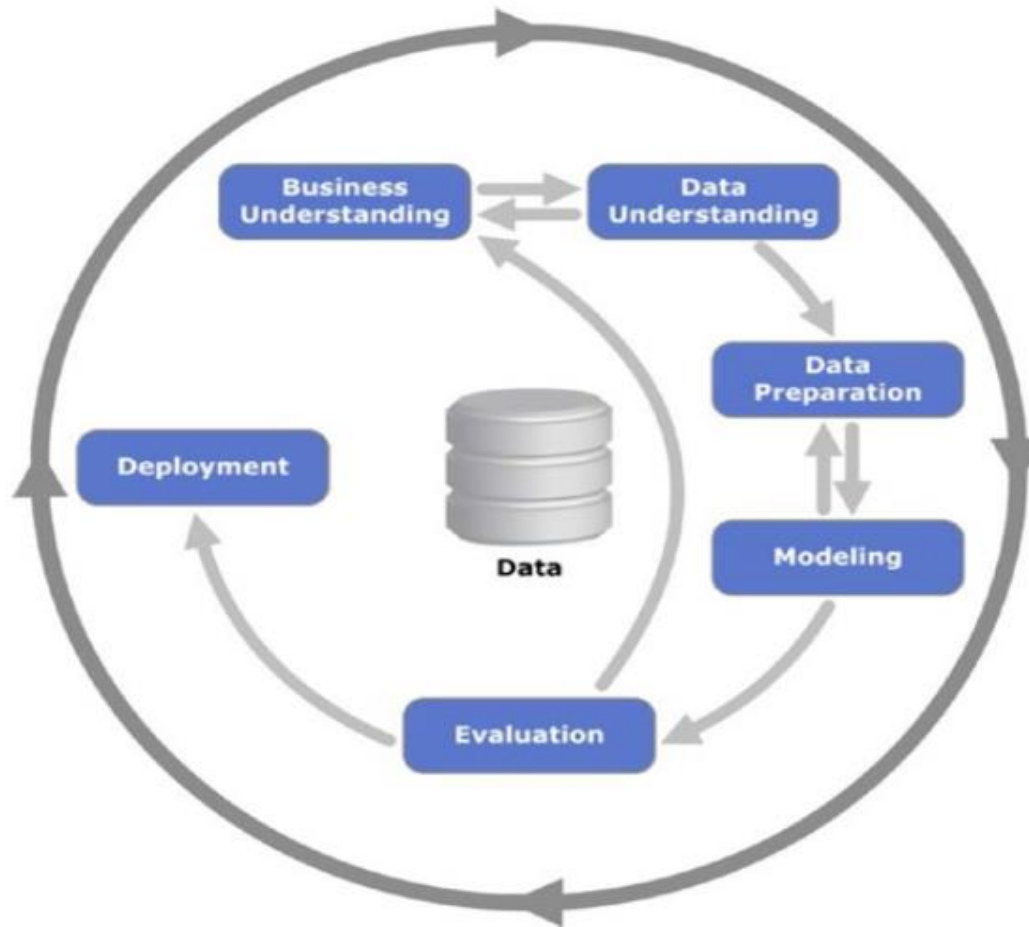
Data Mining Frameworks

- Knowledge Discovery Database Process Model (KDD)
- Cross Industrial Standard Process for Data Mining (CRISP-DM)
- Sample, Explore, Modify, Model and Assess (SEMMA)

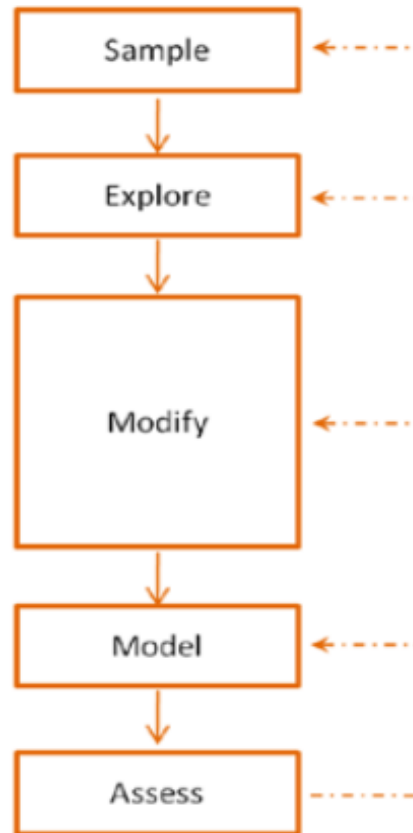
Knowledge Discovery Database Process Model (KDD)



Cross Industrial Standard Process for Data Mining (CRISP-DM)



Sample, Explore, Modify, Model and Assess (SEMMA)



Steps of a KDD Process

- Learning the application domain
 - relevant prior knowledge and goals of application
- Creating a target data set: data selection
- Data cleaning and preprocessing: (may take 60% of effort!)
- Data reduction and transformation
 - Find useful features, dimensionality/variable reduction, invariant representation.
- Choosing functions of data mining
 - summarization, classification, regression, association, clustering.
- Choosing the mining algorithm(s)
- Data mining: search for patterns of interest
- Pattern evaluation and knowledge presentation
 - visualization, transformation, removing redundant patterns, etc.
- Use of discovered knowledge

KDD Process Ex: Web Log

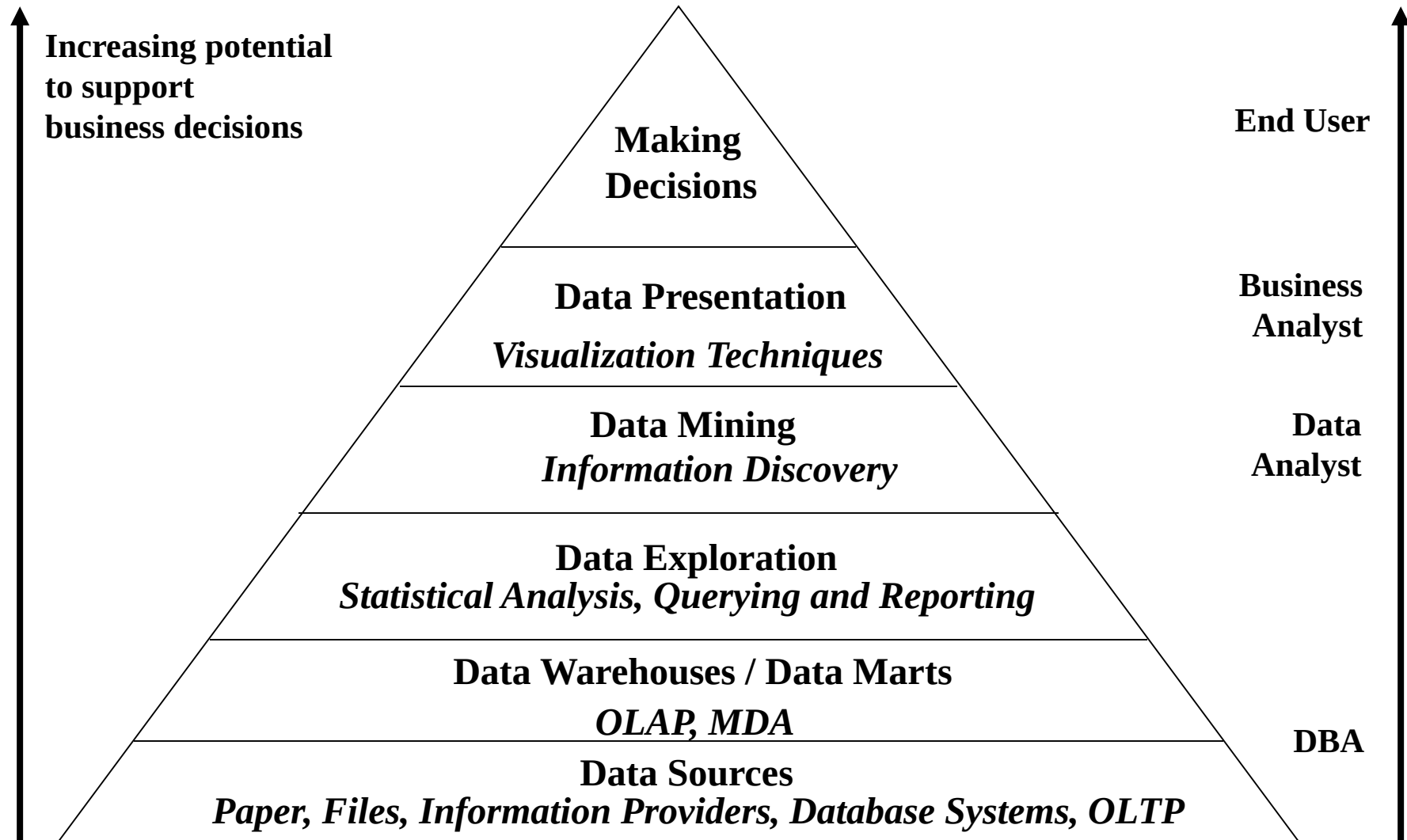
- ***Selection:***
 - Select log data (dates and locations) to use
- ***Preprocessing:***
 - Remove identifying URLs
 - Remove error logs
- ***Transformation:***
 - Sessionize (sort and group)
- ***Data Mining:***
 - Identify and count patterns
 - Construct data structure
- ***Interpretation/Evaluation:***
 - Identify and display frequently accessed sequences.
- ***Potential User Applications:***
 - Cache prediction
 - Personalization

Definition of Data Mining

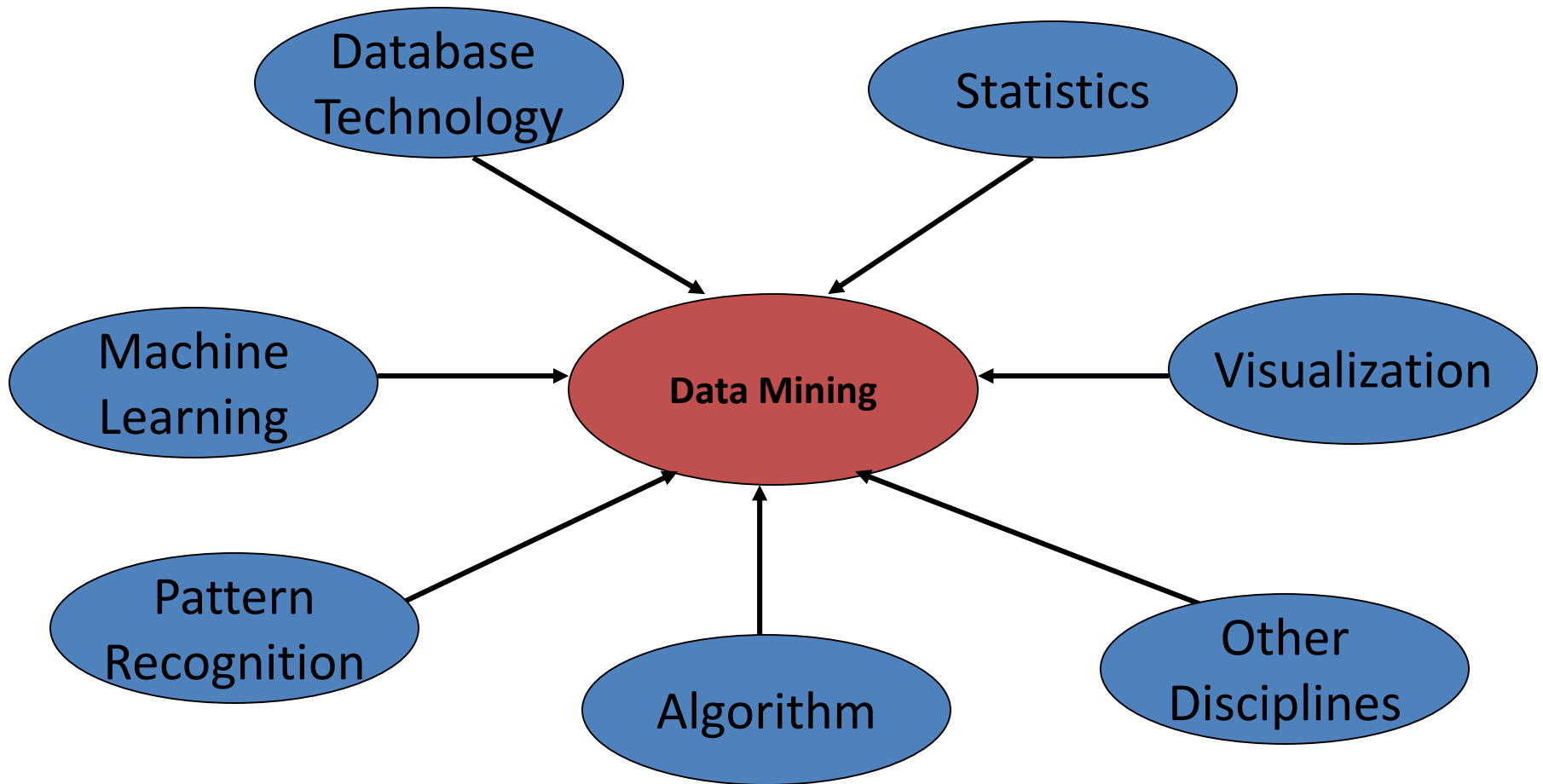


- The non-trivial process of identifying valid, novel, potentially useful, and ultimately understandable patterns in data stored in structured databases.
- *Fayyad et al., (1996)*
- Keywords in this definition: Process, nontrivial, valid, novel, potentially useful, understandable.
- Other names: knowledge extraction, pattern analysis, knowledge discovery, information harvesting, pattern searching, data dredging.

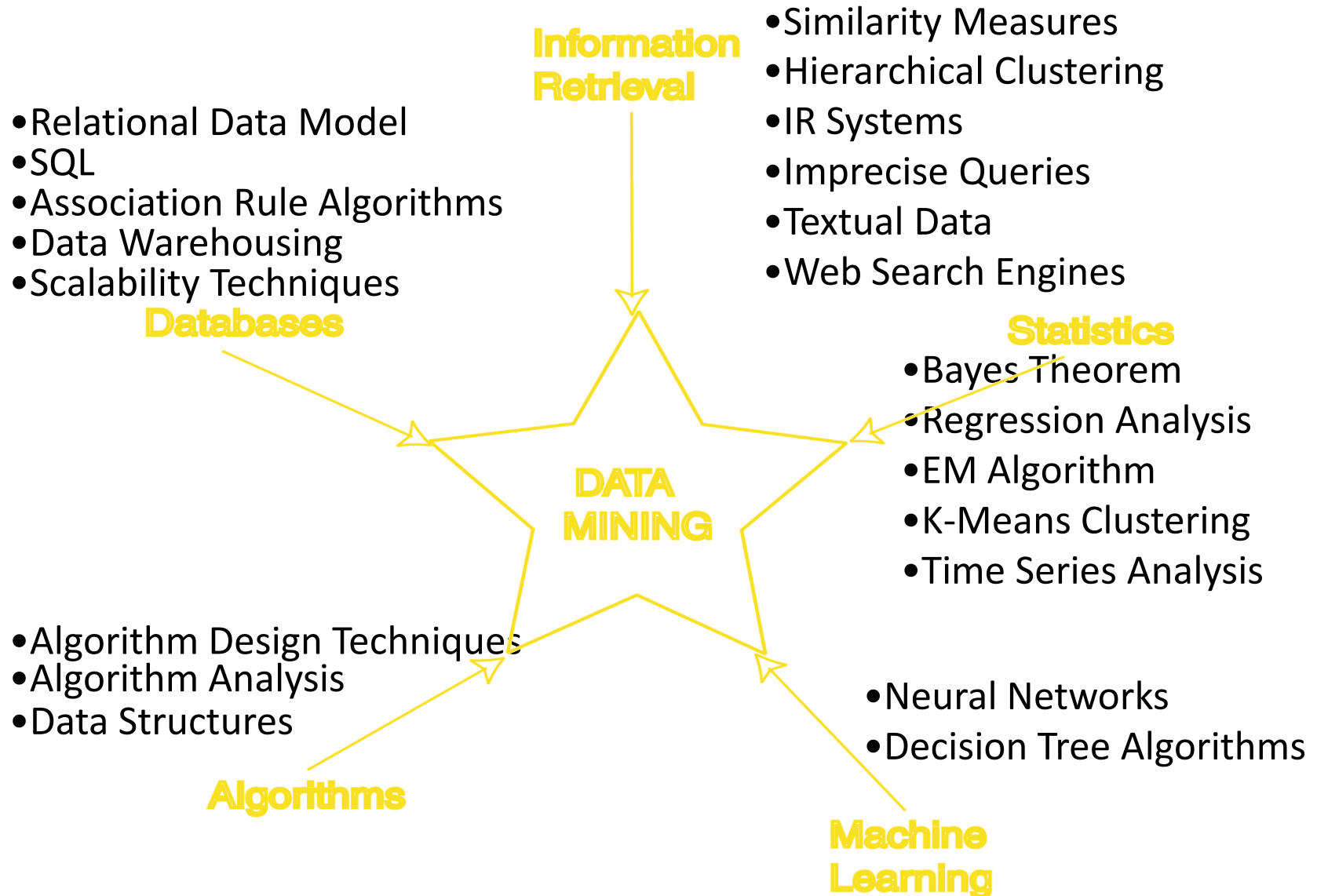
Data Mining and Business Intelligence



Data Mining: Confluence of Multiple Disciplines



Data Mining Development



Database Processing vs. Data Mining Processing

- Query
 - Well defined
 - SQL

- Data
 - Operational data

- Output
 - Precise
 - Subset of database

- Query
 - Poorly defined
 - No precise query language

- Data
 - Not operational data

- Output
 - Fuzzy
 - Not a subset of database

Query Examples

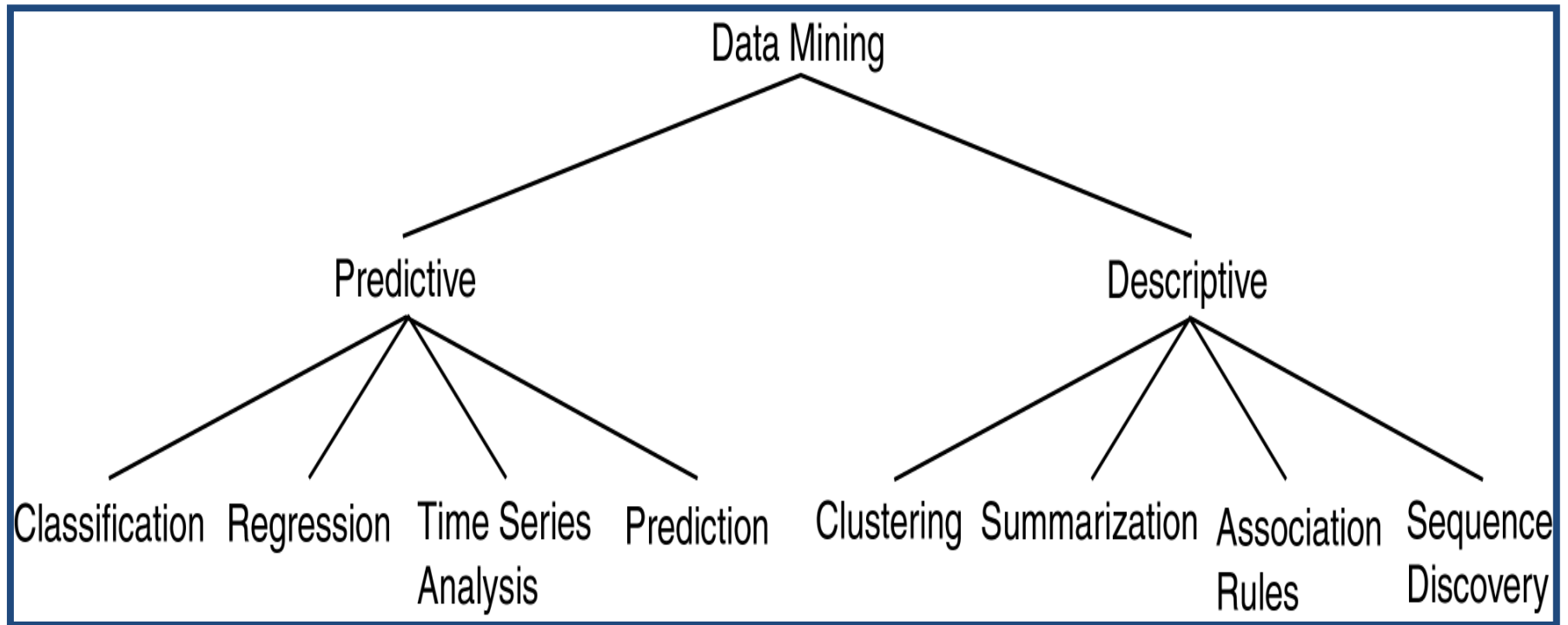
- Database

- Find all credit applicants with last name of Smith.
- Identify customers who have purchased more than \$10,000 in the last month.
- Find all customers who have purchased milk

- Data Mining

- Find all credit applicants who are poor credit risks. (classification)
- Identify customers with similar buying habits. (Clustering)
- Find all items which are frequently purchased with milk. (association rules)

Data Mining Models and Tasks

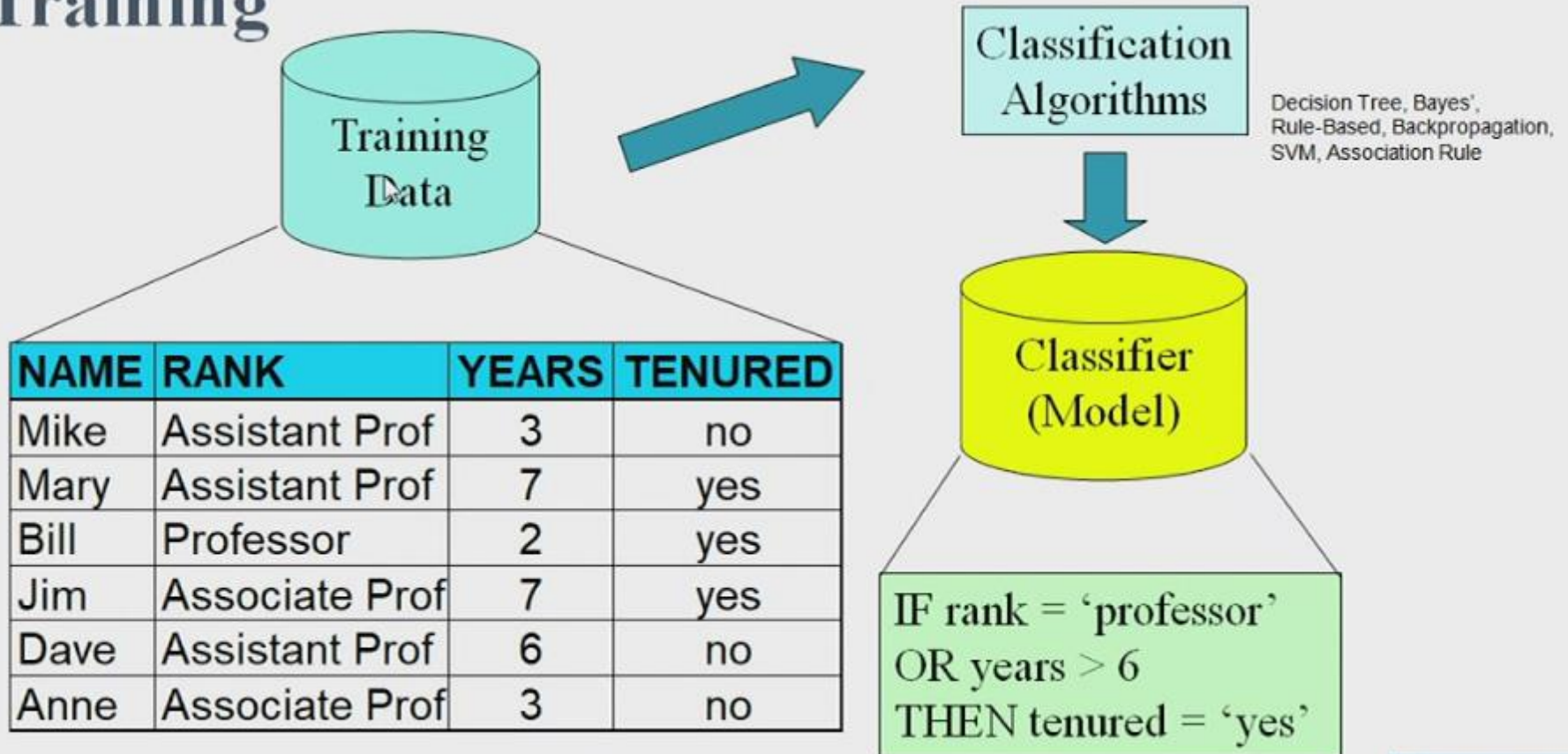


Classification Vs. Regression

Classification	Regression
• Classification is the task of predicting a discrete class label • In a classification problem data is labelled into one of two or more classes • A classification problem with two classes is called binary, more than two classes is called a multi-class classification • Classifying an email as spam or non-spam is an example of a classification problem	• Regression is the task of predicting a continuous quantity • A regression problem requires the prediction of a quantity • A regression problem with multiple input variables is called a multivariate regression problem • Predicting the price of a stock over a period of time is a regression problem

Classification

Training

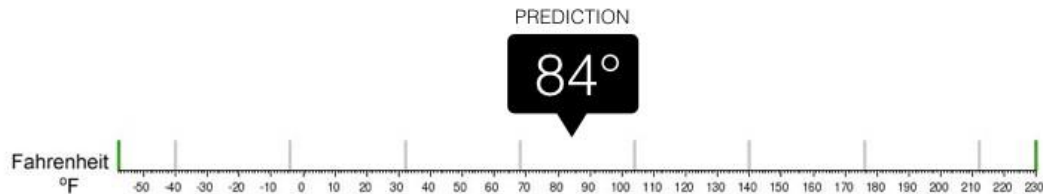


Classification Vs. Prediction



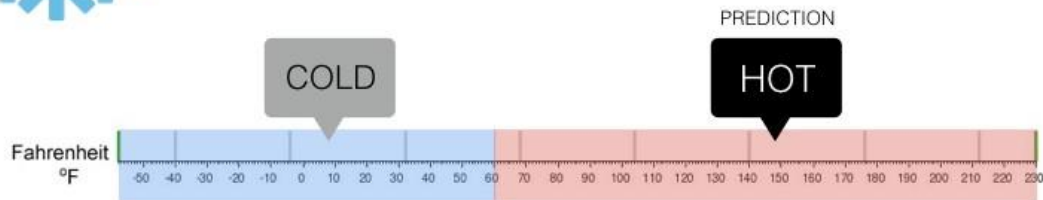
Regression

What is the temperature going to be tomorrow?

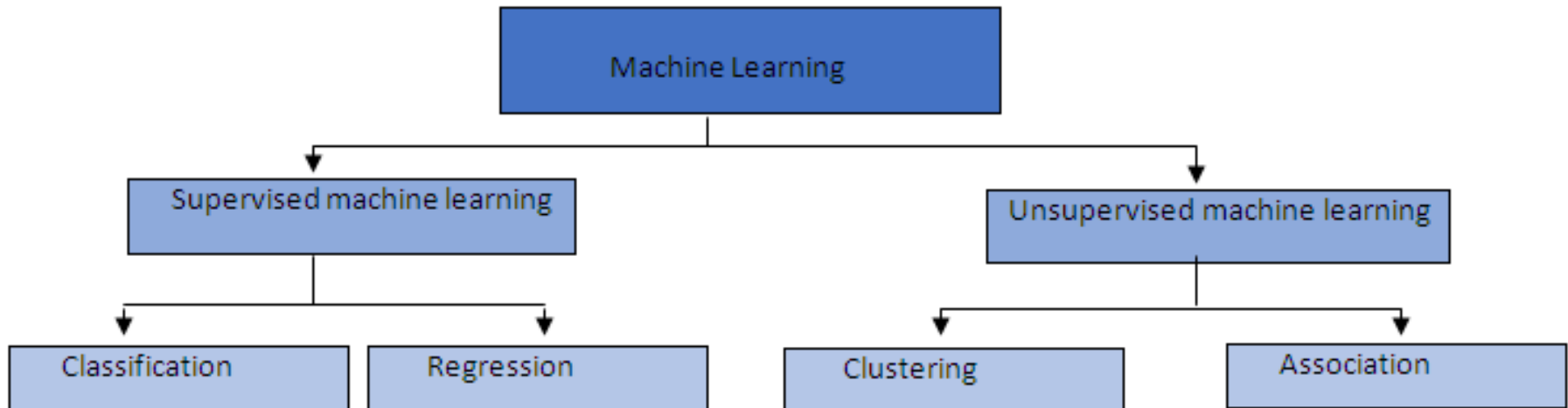


Classification

Will it be Cold or Hot tomorrow?



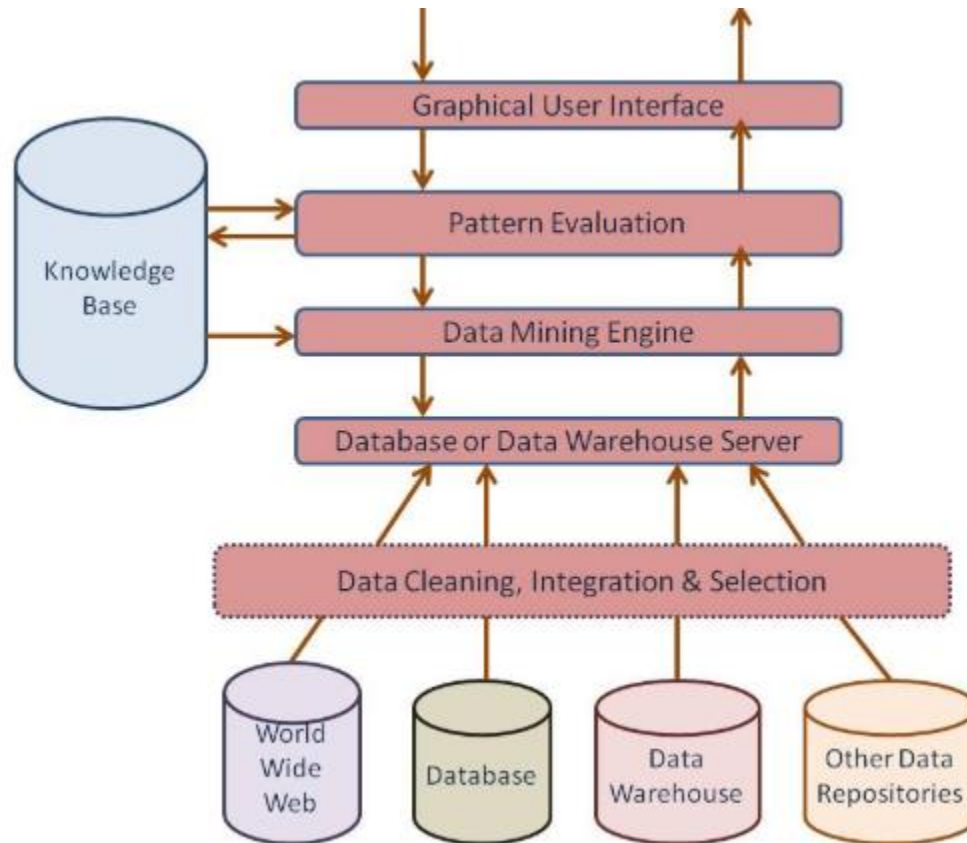
From the Perspective of Machine Learning



Supervised Learning Vs. Unsupervised Learning

Supervised Classification	Unsupervised Clustering
• known number of classes • based on a training set • used to classify future observations	• unknown number of classes • no prior knowledge • used to understand (explore) data

Architecture of Data Mining System



Data Mining Tasks

- **Classification** maps data into predefined groups or classes
 - Supervised learning
 - Pattern recognition
 - Prediction
- **Regression** is used to map a data item to a real valued prediction variable.
- **Clustering** groups similar data together into clusters.
 - Unsupervised learning
 - Segmentation
 - Partitioning

Basic Data Mining Tasks (cont'd)

- **Summarization** maps data into subsets with associated simple descriptions.
 - Characterization
 - Generalization
- **Link Analysis** uncovers relationships among data.
 - Affinity Analysis
 - Association Rules
 - Sequential Analysis determines sequential patterns.

Multi-Dimensional View of Data Mining

- **Data to be mined**
 - Relational, data warehouse, transactional, stream, object-oriented/relational, active, spatial, time-series, text, multi-media, heterogeneous, legacy, WWW
- **Knowledge to be mined**
 - Characterization, discrimination, association, classification, clustering, trend/deviation, outlier analysis, etc.
 - Multiple/integrated functions and mining at multiple levels
- **Techniques utilized**
 - Database-oriented, data warehouse (OLAP), machine learning, statistics, visualization, etc.
- **Applications adapted**
 - Retail, telecommunication, banking, fraud analysis, bio-data mining, stock market analysis, text mining, Web mining, etc.

Data Mining: Classification Schemes

- General functionality
 - Descriptive data mining
 - Predictive data mining
- Different views lead to different classifications
 - **Data** view: Kinds of data to be mined
 - **Knowledge** view: Kinds of knowledge to be discovered
 - **Method** view: Kinds of techniques utilized
 - **Application** view: Kinds of applications adapted

Data Mining: On What Kinds of Data?

- Database-oriented data sets and applications
 - Relational database, data warehouse, transactional database
- Advanced data sets and advanced applications
 - Data streams and sensor data
 - Time-series data, temporal data, sequence data (incl. bio-sequences)
 - Structure data, graphs, social networks and multi-linked data
 - Object-relational databases
 - Heterogeneous databases and legacy databases
 - Spatial data and spatiotemporal data
 - Multimedia database
 - Text databases
 - The World-Wide Web

Data Mining Functionalities

- Multidimensional concept description: Characterization and discrimination
 - Generalize, summarize, and contrast data characteristics, e.g., dry vs. wet regions
- Frequent patterns, association, correlation vs. causality
 - Diaper → Beer [0.5%, 75%] (Correlation or causality?)
- Classification and prediction
 - Construct models (functions) that describe and distinguish classes or concepts for future prediction
 - E.g., classify countries based on (climate), or classify cars based on (gas mileage)
 - Predict some unknown or missing numerical values

Data Mining Functionalities (Cont...)

- Cluster analysis
 - Class label is unknown: Group data to form new classes, e.g., cluster houses to find distribution patterns
 - Maximizing intra-class similarity & minimizing interclass similarity
- Outlier analysis
 - Outlier: Data object that does not comply with the general behavior of the data
 - Noise or exception? Useful in fraud detection, rare events analysis
- Trend and evolution analysis
 - Trend and deviation: e.g., regression analysis
 - Sequential pattern mining: e.g., digital camera → large SD memory
 - Periodicity analysis
 - Similarity-based analysis
- Other pattern-directed or statistical analyses

Major Issues in Data Mining

- Mining methodology
 - Mining different kinds of knowledge from diverse data types, e.g., bio, stream, Web
 - Performance: efficiency, effectiveness, and scalability
 - Pattern evaluation: the interestingness problem
 - Incorporation of background knowledge
 - Handling noise and incomplete data
 - Parallel, distributed and incremental mining methods
 - Integration of the discovered knowledge with existing one: knowledge fusion
- User interaction
 - Data mining query languages and ad-hoc mining
 - Expression and visualization of data mining results
 - Interactive mining of knowledge at multiple levels of abstraction
- Applications and social impacts
 - Domain-specific data mining & invisible data mining
 - Protection of data security, integrity, and privacy

Summary

- Data mining: Discovering interesting patterns from large amounts of data
- A natural evolution of database technology, in great demand, with wide applications
- A KDD process includes data cleaning, data integration, data selection, transformation, data mining, pattern evaluation, and knowledge presentation
- Mining can be performed in a variety of information repositories
- Data mining functionalities: characterization, discrimination, association, classification, clustering, outlier and trend analysis, etc.
- Data mining systems and architectures
- Major issues in data mining